

Question 1

- (a) False. Consider metric space $(\mathbb{R}, d_E)$. A set $\{\frac{1}{n}\}$ for some $n \in \mathbb{N}$ is complete as every sequence in $\{\frac{1}{n}\}$ is constant and thus converges to $\frac{1}{n}$ and so every Cauchy sequence converges to $\frac{1}{n}$ and so $\{\frac{1}{n}\}$ is complete. But now $\bigcup_{n \in \mathbb{N}} \{\frac{1}{n}\}$ is not complete, as Cauchy sequence $(\frac{1}{n})$ does not converge in $\bigcup_{n \in \mathbb{N}} \{\frac{1}{n}\}$ (since it converges to 0 in metric space $(\mathbb{R}, d_E)$ but $0 \notin \bigcup_{n \in \mathbb{N}} \{\frac{1}{n}\}$) and is thus not complete.
- (b) True. Let (x_n) be a Cauchy sequence in $\bigcap_{i \in I} A_i$. Then $x_n \in A_i$ for all $n \in \mathbb{N}$ and for all $i \in I$. But then (x_n) is Cauchy in A_i and since A_i is complete, x_n converges to some $x \in A_i$. Since this is true for all $i \in I$, we get $x \in \bigcap_{i \in I} A_i$ and so $\bigcap_{i \in I} A_i$ is complete.
- (c) False. Consider metric spaces $(\mathbb{R}, d_D)$ and $(\mathbb{R}, d_E)$ and continuous function $f : (\mathbb{R}, d_D) \rightarrow (\mathbb{R}, d_E)$ defined as $f(x) = x$. This is continuous because f maps from a set equipped with the discrete metric. Now consider $(0, 1]$. Under d_D , a sequence (x_n) in $(0, 1]$ is Cauchy if and only if it is eventually constant and thus $(0, 1]$ is complete. But then $f((0, 1]) = (0, 1]$ which is not complete under $(\mathbb{R}, d_E)$ by considering Cauchy sequence $(\frac{1}{n})$ which converges to $0 \notin (0, 1]$.
- (d) True. Let (x_n) be a Cauchy sequence in $f^{-1}(B)$. Then $(f(x_n))$ is a sequence in B . Since (X, d_X) is complete, $x_n \rightarrow x$ for some $x \in X$. Then because f is continuous, $f(x_n) \rightarrow f(x)$ for $f(x) \in Y$. Then $(f(x_n))$ is convergent, and so is Cauchy. Then because B is complete, $f(x) \in B$ which implies that $x \in f^{-1}(B)$ and so $f^{-1}(B)$ is a complete subset of X .
- (e) False. Consider metric space $(\mathbb{R}, d_D)$. Since every Cauchy sequence is eventually constant, $(\mathbb{R}, d_D)$ is complete. Now let $A_n = (0, \frac{1}{n})$. Then for $m > n$, $\frac{1}{m} < \frac{1}{n}$ and so $(0, \frac{1}{m}) \subset (0, \frac{1}{n})$ but $(0, \frac{1}{m}) \neq (0, \frac{1}{n})$ as, for example, $\frac{1}{m}$ is in $(0, \frac{1}{n})$ but not $(0, \frac{1}{m})$. So $(A_n)_{n \in \mathbb{N}}$ is a strictly decreasing sequence of subsets of X . Now $\overline{A_n} = A_n$ as every set is closed under d_D . But then $\bigcap_{n \in \mathbb{N}} \overline{A_n} = \bigcap_{n \in \mathbb{N}} (0, \frac{1}{n}) = \emptyset$.

To show this, suppose $x \in \bigcap_{n \in \mathbb{N}} (0, \frac{1}{n})$. Then since $x \in A_1 = (0, 1)$, we get $x > 0$.

But then by the Archimedean property there exists some $n \in \mathbb{N}$ such that $\frac{1}{n} < x$ so $x \notin (0, \frac{1}{n})$ which is a contradiction. Thus there cannot be any $x \in \bigcap_{n \in \mathbb{N}} (0, \frac{1}{n})$ and so

$$\bigcap_{n \in \mathbb{N}} (0, \frac{1}{n}) = \emptyset$$

Question 2

- (a) Notice if $y \in Y$, then since X is dense in Y , we get $\overline{X} = Y$ and so there exists a sequence (x_n) in X such that $x_n \rightarrow y$.

So define $g : Y \rightarrow Z$ as: $g(x) = \lim_{n \rightarrow \infty} (f(x_n))$

We first show g is well-defined. Let (x_n) and (x'_n) be sequences in X such that $x_n \rightarrow y$ and $x'_n \rightarrow y$. Then because f is continuous, $f(x_n) \rightarrow f(y)$ and $f(x'_n) \rightarrow f(y)$ and because (Z, d_Z) is complete, $f(y) \in Z$ and thus g is well defined.

Now if $x \in X$, then the sequence (x_n) converges to x and so $g(x) = \lim_{n \rightarrow \infty} (f(x_n)) = f(x)$ since f is continuous.

Now we show g is uniformly continuous. Let $\epsilon > 0$. Since f is uniformly continuous, there exists some $\delta_f > 0$ such that for all $x, x' \in X$, $d_X(x, x') < \delta_f \Rightarrow d_Z(f(x), f(x')) < \frac{\epsilon}{2}$. Let $\delta = \frac{\delta_f}{3}$ and suppose that $y, y' \in Y$ such that $d_Y(y, y') < \delta$. Then there exists sequences (x_n) and (x'_n) in X such that $x_n \rightarrow y$ and $x'_n \rightarrow y'$ under d_Y . Now $d_X(x_n, x'_n) = d_Y(x_n, x'_n)$ since (X, d_X) is a subspace of (Y, d_Y) . Then, by the Triangle Inequality, $d_Y(x_n, x'_n) \leq d_Y(x_n, y) + d_Y(y, y') + d_Y(y', x'_n)$. Because $x_n \rightarrow y$ and $x'_n \rightarrow y'$, there is some $N \in \mathbb{N}$ such that for all $n \geq N$, $d_Y(x_n, y) < \delta$ and $d_Y(y', x'_n) < \delta$. So $d_X(x_n, x'_n) = d_Y(x_n, x'_n) < 3\delta = \delta_f$. Thus $d_Z(f(x_n), f(x'_n)) < \frac{\epsilon}{2}$. Now $d_Z(g(y), g(y')) = d_Z(\lim_{n \rightarrow \infty} (f(x_n)), \lim_{n \rightarrow \infty} (f(x'_n))) = \lim_{n \rightarrow \infty} d_Z(f(x_n), f(x'_n))$, by continuity of a metric. Then $\lim_{n \rightarrow \infty} d_Z(f(x_n), f(x'_n)) \leq \frac{\epsilon}{2} < \epsilon$ as required.

- (b) Consider $X = (\mathbb{Q}, d_E), Y = (\mathbb{R}, d_E), Z = (\mathbb{Q}, d_E)$. Then $f : (\mathbb{Q}, d_E) \rightarrow (\mathbb{Q}, d_E)$ defined by $f(x) = x$ is uniformly continuous. If we define $g : (\mathbb{R}, d_E) \rightarrow (\mathbb{Q}, d_E)$ as above, it will not be well-defined. Consider $e \in \mathbb{R}$. A sequence $((1 + \frac{1}{n})^n)$ converges to e , but then $f((1 + \frac{1}{n})^n) = (1 + \frac{1}{n})^n$ and since $e \notin \mathbb{Q}$, $g(e)$ is not defined and thus g cannot be a valid function